

Curriculum Vitae

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Professional Objective

To develop expertise in experimental Astroparticle Physics through cutting-edge research at world-class laboratories, while contributing to the advancement of ultra-high energy neutrino detection technologies.

Education

Ph.D. in Physics <i>2020–Present</i> University of Kansas, USA	3.98/4.0
M.S. in Electrical Engineering <i>2024–2025</i> University of Kansas, USA	3.87/4.0
M.Sc. in Physics <i>2018–2020</i> Aligarh Muslim University, India	9.170/10 (Gold Medalist)
B.Sc. Honors in Physics <i>2015–2018</i> Aligarh Muslim University, India	9.186/10 (Gold Medalist)
Higher Secondary (12th) <i>2014</i> Central Board of Secondary Education, India	9.120/10 (First Position)
Secondary (10th) <i>2012</i> Central Board of Secondary Education, India	10/10 (Certificate of Merit)

Research Experience

- Primary Analyzer, Askaryan Radio Array (ARA) Station 1** *2021–Present*
Conducting data analysis of ARA Station 1 for ultra-high energy (UHE) neutrino detection with Prof. D. Besson at University of Kansas (KU).
- Analysis Lead, Radio Background Characterization with ARA** *Dec 2025–Present*
Conducting data analysis on “Characterizing Astrophysical and Environmental Phenomena with the Askaryan Radio Array” with Prof. D. Besson at University of Kansas (KU).
- EECS Master’s Project Research** *2024–2025*
Developed **AAFIYA**: **A**ntenna **A**nalysis in **F**requency-domain for **I**mpedance and **Y**ield **A**ssessment, a modular Python toolkit for rigorous, reproducible analysis and visualization of antenna S-parameters, impedance, and gain/beam patterns from both measurement and simulation data, with Prof. Jim Stiles at EECS Department, KU, (Report), (Slides).
- Machine Learning Project** *2025*
Developed a CNN model for “Automatic Surface Detection in Echograms using Deep Learning” as a part of Machine Learning class at EECS Department, KU, (Project Repository), (Report).
- IceCube Neutrino Observatory Monitoring** *2021–2025*
Served as KU representative for monitoring shifts, presenting weekly reports to the collaboration.

Physics Master's Thesis Research 2019–2020

Investigated light sterile neutrino oscillations with Prof. M.S. Athar at Aligarh Muslim University (AMU), producing review report “Light Sterile Neutrinos: Status and Prospects” (DOI:10.13140/RG.2.2.13777.66405).

Summer Research Program May–Jul 2019

Studied the role of θ_{13} mixing angle in three-flavor neutrino oscillations with Prof. S.K. Agarwalla at Institute of Physics, India, producing technical report “Importance of 1-3 Mixing Angle in Three-Flavor Neutrino Oscillations Paradigm” (DOI: 10.13140/RG.2.2.34421.12009).

Winter Internship Program Dec 2019–Jan 2020

Researched “Weak Interaction Physics” at IIT-BHU with Dr. G. Abbas.

Physics Undergraduate Project Research 2017–2018

“Quarks & Leptons”: Detailed study of Standard Model interactions with Prof. M.S. Athar at AMU [View Handwritten Report].

Field Experience

NSF's Summit Station, Greenland Field Campaign Apr–May 2025

Participated in the Radio Neutrino Observatory – Greenland (RNO-G) calibration campaign, performing in-situ radio-frequency measurements and equipment deployment at Summit Station, Greenland.

Publications

First-Author Works

- **Seikh, M.F.H. & Besson, D.** (2026). “Polar Ice as a Laboratory for Physics”. *Page 27-37, Physics Bulletin 2026, Physics Department, AMU*
- **Seikh, M.F.H.** (2025). “A geometrical approach to neutrino oscillation parameters”. *AIP Advances 1 October 2025; 15 (10): 105204*. DOI:10.1063/5.0292233.
- **Seikh, M.F.H., Grachev, V., Kravchenko, I., Athar, M.S., & Besson, D.** 2025. “First Radio Neutrino & Cosmic Rays Astronomy Workshop in India”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2509.20704 DOI:10.1140/epjs/s11734-025-01955-8.
- **Seikh, M.F.H., Giri, P. & Besson, D. for the ARA Collaboration** 2025. “Techniques for Continuous Wave Identification and Filtering in the Askaryan Radio Array”. *PoS(ICRC2025)1171*. arXiv:2509.20735.
- **Seikh, M.F.H. & Besson, D. for the RNO-G Collaboration** 2025. “Observation of Radio Solar Flares in RNO-G”. *PoS(ICRC2025)1365*.
- **Seikh, M.F.H.** (2024). “Physics of radio antennas”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2411.07507 DOI: 10.1140/epjs/s11734-025-01500-7
- **Seikh, M.F.H. for the ARA Collaboration** (2024). “Askaryan Radio Array: searching for the highest energy neutrinos”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2411.01761 DOI: 10.1140/epjs/s11734-025-01708-7
- **Seikh, M.F.H. & Besson, D.** (2024). “IceCube Neutrino Observatory: A Cold Discovery Hub”. *Page 14-23, Physics Bulletin 2024, Physics Department, AMU*
- **Seikh, M.F.H. for the ARA Collaboration** (2023). “Calibration and Physics with ARA Station 1”. *PoS(ICRC2023)1163*. arXiv:2308.07292
- **Seikh, M.F.H.** (2020). “Neutrinos and the 40 Future Routes”. *Physics Bulletin, Physics Department, AMU*. DOI: 10.13140/RG.2.2.27981.10729

Collaborative Works

Full publication (includes contributions to ARA, RNO-G, ARIANNA, IceCube, IceCube Gen2, PUEO, TAROE-M and RET collaborations) record available on:

Inspire HEP ; Google Scholar ; ResearchGate ; ORCID  ; Scopus

Professional Service

Editorial

- **Book Editor** for Springer’s EPJ-ST journal: “Radio Detection of Ultra-high Energy Neutrino & Cosmic Ray” (15 articles) with M.S. Athar, D. Besson, V. Grachev, and I. Karvchenko (2024–2025)
- **Founder and Editor-in-Chief** of *The Magnetic Monopole* e-magazine for KU’s Society of Physics Students [View Issue] (June 2022)

Research Article Review

- **Peer Reviewer:** for World Scientific’s IJMP-A journal (1 article, 2 reviews) (2025)
- **Peer Reviewer:** for Springer’s EPJ-ST journal (5 articles, 11 reviews) (2024–2025)
- **Internal ARA Collaboration Reviewer:** Pawan Giri for ARA Collaboration, “A Comprehensive Diffuse Neutrino Search Using the Full Askaryan Radio Array”, 32nd International Symposium on Lepton Photon Interactions at High Energies (November 2025)

Other Reviews

- **REU Application Reviewer:** for KU Physics and Astronomy (10 Applications) (2025)
- **SSGSA Application Reviewer:** for SSGSA (Multiple Applications) (2021, 2022, 2023, 2024, 2025)

Professional Leadership/Membership

- **American Physical Society (APS):** Graduate Student Member (2020–Present)
- **National Society of Black Physicists:** Student Member (2021–Present)
- **Physics & Astronomy Graduate Student Organization (GSO), KU:**
 - ✓ Graduate Students Representative - Member at Large (2025–Present)
 - ✓ Graduate Students Representative - Liaison (2024–2025)
 - ✓ GSO Executive Officer (2023–2024)
- **Sir Syed Global Scholar Award (SSGSA):**
 - ✓ Peer Mentor (2023–Present)
 - ✓ Team Lead, Promotions & Outreach (2023–2025)
 - ✓ Member, Mentorship & Applications Team (2021–2023)
 - ✓ AMU Campus Team Member (2019–2020)
- **Society of Physics Students, KU Chapter:** Graduate President (2021–2022)
- **Sigma Pi Sigma (National Physics Honor Society):** Lifetime Member
- **AMU Society of Physics:** Member (2018–Present)

Technical Proficiencies

- **Programming Languages:** C/C++, Python, FORTRAN
- **Physics Software:** CERN ROOT, AraRoot, Wolfram Mathematica, HFSS

- **Operating Systems:** Linux (Ubuntu), Windows, macOS
- **Document Preparation:** L^AT_EX, Microsoft Office Suite

Honors & Awards

Academic Honors

- **University Gold Medals:**
 - ✓ M.Sc. Physics (Top among 50 students), AMU (*2020*)
 - ✓ B.Sc. Honors Physics (Top among 120 students), AMU (*2018*)
 - ✓ Highest percentage in B.Sc. (Hons) Faculty of Science, AMU (*2018*)
- **Abdul Aziz Medal** for B.Sc. Physics, AMU (*2018*)
- **Undergraduate Topper Scholarship**, UGC India (*2020*)
- **University Merit Financial Award**, AMU (*2019–2020*)
- **First Position** in Higher Secondary (12th) among 40 students (*2014*)
- **Certificate of Merit** for perfect 10/10 CGPA in Secondary (10th) (*2012*)

Fellowships & Grants

- **Ralston Dissertation Fellowship**, KU Physics (*2025–2026*)
- **McKeithan Summer Fellowship**, KU Physics (*2025*)
- **Graduate Student Award for Distinguished Service**, KU (*2024–2025*)
- **Wallace S. Strobel Scholarship**, KU EECS (*2024–2025*)
- **Ralston Summer Fellowship**, KU Physics (*2024*)
- **Doctoral Student Research Fund**, KU (*2024*)
- **Distinguished Students Award**, APS FIP (*2021–2022*)
- **Carl & Catherine Chaffee Scholarship**, KU Physics (*2020–2021*)

Conference Travel Awards

- **Graduate Student Travel Fund**, KU Physics (*2023, 2025*)
- **DAP Travel Grant**, APS (*2024*)
- **DPF Travel Award**, APS (*2022, 2024*)
- **CLAS Graduate Scholarly Development Fund**, KU (*2022*)

Other Distinctions

- **Washington DC Scholarship**, AMU Alumni Association (*2015–2019*)
- **Sir Syed Global Scholar Award** for prospective graduate students (*2018*)

Invited Lectures, Seminars, & Colloquia

- **Radio Engineering for Particle Astrophysics Experiments**
HEP Seminar, University of Nebraska-Lincoln (*March 26, 2025*)
- **Radio Engineering for Particle Astrophysics Experiments**
Astroparticle Seminar, Ohio State University (*January 22, 2025*)
- **Radio Engineering & Antenna Physics**
Lecture Series, IEI Summer School 2024, University of Alaska, Anchorage (*July 2–5, 2024*)
- **UHE Neutrino & Cosmic Ray Detectors**
ICEHAP Seminar, Chiba University, Japan (*August 9, 2023*)

- **Antenna Physics & Measurements**
Lecture Series, IEI Summer School 2023, South Dakota School of Mines (*July 2–5, 2023*)
- **UHE Neutrino & Cosmic Ray Detectors**
Physics Colloquium, AMU, India (*January 12, 2023*)
- **Antennas in UHE Neutrino Detectors**
Particle Physics Seminar, University of Delaware (*October 4, 2022*)

Outreach & Leadership

- Co-organized and hosted India’s inaugural workshop on **Radio Detection of UHE Neutrino and Cosmic Rays** at AMU (*April 2024*)
- Conducted outreach seminar on **Physics Research Opportunities** at AMU (*2021*)
- Participated in India’s first **Science Leadership Workshop** (*2020*)
- Volunteer for **One Person One Tree** plantation drive (*2017*)
- Participant in **Blood Donation Camp** (*2017*)

Conference Presentations

Oral Presentations

- **RNO-G in-ice Antennas**
RNO-G Collaboration Meeting, Uppsala University (*September 9–14, 2024*)
- **Radio Frequency Instrumentation**
Radio Neutrino Workshop, AMU, India (*April 18–20, 2024*)
- **Towards Five Station ARA Analysis**
APS April Meeting, Sacramento (*April 3–6, 2024*)
- **ARA Antenna Models**
ARA Collaboration Meeting, Ohio State University (*March 5–8, 2024*)
- **Current Status & Future of ARA**
APS Prairie Section Meeting, University of Missouri (*November 30–December 2, 2023*)
- **Introduction to Antenna Physics**
RET Collaboration Meeting, Vrije University Brussels (*November 8–11, 2023*)
- **Gain Calibration of RNO-G Antennas**
RNO-G Fall Meeting, Penn State University (*September 11–15, 2023*)
- **Galactic Noise in Radio Neutrino Experiments**
Astro Seminar, KU (*October 28, 2022*)
- **Towards ARA Station 1 Calibration**
APS April Meeting, New York City (*April 9–12, 2022*)
- **RNO-G Ice Calibration Planning**
RNO-G Collaboration Meeting, University of Chicago (*April 8, 2022*)

Poster Presentations

- **CW Identification & Filtering Techniques in ARA**
39th ICRC, Geneva, Switzerland (*July 19, 2025*)
- **Observation of radio solar flares in RNO-G**
39th ICRC, Geneva, Switzerland (*July 15, 2025*)
- **Gain Measurements of LPDA Antennas**
APS April Meeting, Sacramento (*April 3–6, 2024*)

- **Overview of ARA Detector: Five Station Analysis**
ICTS Neutrino School, India (*April 22–May 3, 2024*)
- **Calibration of ARA Station 1**
38th ICRC, Nagoya University, Japan (*July 29, 2023*)

Professional Development

- **Responsible Conduct of Research**, (CITI Program, *2021, 2025*) - 100%
- **From the Big Bang to Dark Energy** (U Tokyo Coursera, *2020*) - 100%
- **Particle Physics: Introduction** (U Geneva Coursera, *2019*) - 100%
- **Senior Diploma in Painting** (Bangiya Sangeet Parishad) - Distinction

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Special Topics

Radio Detection of Ultra-high Energy Neutrino and Cosmic Ray

Mohammad Ful Hossain Seikh, Mohammad Sajjad Athar,
Dave Besson, Victor Grachev, and Ilya Kravchenko (Eds.)



Cosmic rays and neutrinos: exploring high-energy messengers through radio detection.
Courtesy of Mohammad Ful Hossain Seikh

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